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 Studierichting: Informatica
 Oefeningenreeks 3A nr. 40.7

$$f(x) = \frac{\sqrt{2x^3 - 2x + 1}}{\sqrt[3]{x + \frac{1}{x}}}$$

$$f'(x) = \frac{(\sqrt{2x^3 - 2x + 1})' \cdot \sqrt[3]{x + \frac{1}{x}} - \sqrt{2x^3 - 2x + 1} \cdot \left(\sqrt[3]{x + \frac{1}{x}}\right)'}{\left(\sqrt[3]{x + \frac{1}{x}}\right)^2}$$

$$(\sqrt{2x^3 - 2x + 1})' = \frac{1}{2\sqrt{2x^3 - 2x + 1}} \cdot (6x^2 - 2) = \frac{3x^2 - 1}{\sqrt{2x^3 - 2x + 1}}$$

$$\left(\sqrt[3]{x + \frac{1}{x}}\right)' = \frac{1}{3\left(x + \frac{1}{x}\right)^{\frac{2}{3}}} \cdot \left(1 - \frac{1}{x^2}\right) = \frac{x^2 - 1}{3x^2 \sqrt[3]{\left(x + \frac{1}{x}\right)^2}}$$

$$f'(x) = \frac{\frac{3x^2 - 1}{\sqrt{2x^3 - 2x + 1}} \cdot \sqrt[3]{x + \frac{1}{x}} - \sqrt{2x^3 - 2x + 1} \cdot \frac{x^2 - 1}{3x^2 \sqrt[3]{\left(x + \frac{1}{x}\right)^2}}}{\sqrt[3]{\left(x + \frac{1}{x}\right)^2}}$$

$$f'(x) = \frac{(3x^2 - 1) \cdot 3x^2 \cdot \left(x + \frac{1}{x}\right) - (2x^3 - 2x + 1)(x^2 - 1)}{3x^2 \sqrt{2x^3 - 2x + 1} \sqrt[3]{\left(x + \frac{1}{x}\right)^2} \cdot \sqrt[3]{\left(x + \frac{1}{x}\right)^2}}$$

$$f'(x) = \frac{(3x^2 - 1) \cdot 3x^2 \cdot \left(\frac{x^2 + 1}{x}\right) - (2x^5 - 2x^3 - 2x^3 + 2x + x^2 - 1)}{3x^2 \sqrt{2x^3 - 2x + 1} \left(x + \frac{1}{x}\right)^{\frac{4}{3}}}$$

$$f'(x) = \frac{3x(3x^2 - 1)(x^2 + 1) - (2x^5 - 4x^3 + x^2 + 2x - 1)}{3x^2 \sqrt{2x^3 - 2x + 1} \sqrt[3]{\left(x + \frac{1}{x}\right)^4}}$$

$$f'(x) = \frac{3x(3x^4 + 2x^2 - 1) - 2x^5 + 4x^3 - x^2 - 2x + 1}{3x^2 \sqrt{2x^3 - 2x + 1} \sqrt[3]{\left(x + \frac{1}{x}\right)^4}}$$

$$f'(x) = \frac{9x^5 + 6x^3 - 3x - 2x^5 + 4x^3 - x^2 - 2x + 1}{3x^2 \sqrt{2x^3 - 2x + 1} \sqrt[3]{\left(x + \frac{1}{x}\right)^4}}$$

$$f'(x) = \frac{7x^5 + 10x^3 - x^2 - 5x + 1}{3x^2\sqrt{2x^3 - 2x + 1}\sqrt[3]{\left(x + \frac{1}{x}\right)^4}}$$

References